



MECHATRONICS
SPECTRUM
TEAM



MAE

Machines 1 Test BANK

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ملاحظات هامة:

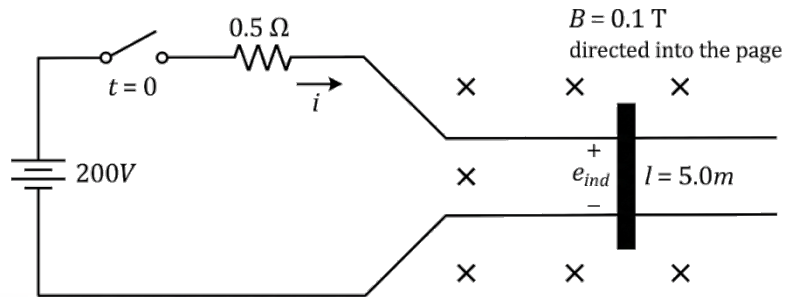
- يمنع نسخ الملف هذا وتغيير شعار Spectrum-Team .
- دوائر ال MID بيحي منها بال FINAL .

الرمز	الدلالة
*	يتكرر بالسنوات لكن تتغير أرقام السؤال
*	يتكرر بالسنوات لكن تتغير كلمات السؤال
*	يتكرر بالسنوات لكن تتغير خيارات السؤال
*	يتكرر بالسنوات كما هو
*	أسئلة سنوات شاشات
*	أسئلة كويات

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Part. 1 (MID MCQs):

* For Q1. to Q5., the non-load linear DC machine shown in the figure:



Q1. The starting current is:

- A) 100 A
- B) 300 A
- C) 400 A
- D) 200 A

Q2. The steady state velocity is:

- A) 10 m/s
- B) 20 m/s
- C) 400 m/s
- D) 200 m/s

Q3. If a 20 N force is opposing the motion of the machine, the steady-state current demanded by the machine is:

- A) 10 A
- B) 40 A
- C) 2 A
- D) 20 A

Q4. If a 20 N force is opposing the motion of the machine, the steady-state velocity of the machine is:

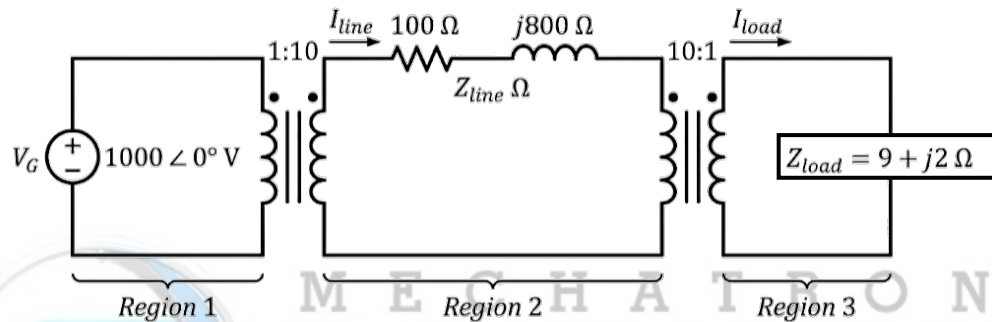
- A) 360 m/s
- B) 120 m/s
- C) 100 m/s
- D) 160 m/s

Q5. If a 20N force is opposing the motion of the machine, the power consumed by the machine at steady-state is:

- A) 1600 W
- B) 3200 W
- C) 4800 W
- D) 2000 W

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For Q6. to Q10, A single phase system shown below contains a 1000 V generator connected to an ideal 1 to 10 step up transformer transmission line $100 + j 800 \Omega$ and 10 to 1 step down transformer and a load The impedance of the transmission line is $100 + j 800 \Omega$ and the impedance of the load is $9 + j 2 \Omega$ The base values for this system are S base 10 kVA and V base at generator side 1000 V:



Q6. The per unit impedance of the line is:

- A) $0.1 + j0.8$ pu
- B) $0.01 + j0.08$ pu
- C) $1.0 + j8.0$ pu
- D) $10 + j80$ pu

Q7. The per unit impedance of the load is:

- A) $0.09 + j0.02$ pu
- B) $0.9 + j0.2$ pu
- C) $9.0 + j2.0$ pu
- D) $90 + j20$ pu

Q8. The current consumed by the load is:

- A) 0.7071 A
- B) 7.071 A
- C) 70.711 A
- D) 707.11 A

Q9. The voltage at the load terminals is:

- A) 65.2 V
- B) 65.2 kV
- C) 6.52 kV
- D) 652 V

Q10. The losses in the transmission line is:

- A) 5 kW
- B) 1.5 kW
- C) 1.25 kW
- D) 0.5 kW

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* For Q11. to Q16., A 30 KVA, 3/0.3 KV, single phase transformer has been tested with Short circuit test. As the following data were taken: $V_{sc} = 160V$, $I_{sc} = 8A$, $P_{sc} = 1024W$

Q11. Find the full load copper losses (P_{cu}).

- A) 1024 W
- B) 1600 W
- C) 1400 W
- D) 1280 W

Q12. Find the magnetic losses (P_{core}).

- A) 1369 W
- B) 1480 W
- C) 1870 W
- D) 876 W

Q13. Find the impedance of the equivalent circuit referred to the secondary side.

- A) 2000Ω
- B) 16Ω
- C) 0.2Ω
- D) 20Ω

Q14. Find the resistance of the equivalent circuit referred to the secondary side.

- A) 16Ω
- B) 0.2Ω
- C) 2Ω
- D) 20Ω

Q15. Find the maximum efficiency occurs at 0.925 full load at 0.9 power factor.

- A) 89.404%
- B) 90.12%
- C) 88.64%
- D) 96.61%

Q16. If the transformer operates at full load and 0.8 power factor lagging. Find the voltage regulation.

- A) 4.8%
- B) 5.2%
- C) 6.4%
- D) 7.1%

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- Q17. *** A transformer with primary winding having 500 turns, secondary winding having 100 turns, if 200 V is applied to the primary winding, the voltage at the secondary winding will be:
- A) 2000 V
 - B) 1000 V
 - C) 500 V
 - D) None of the above
- Q18. *** A transformer with primary winding having 100 turns, secondary winding having 500 turns, if 200V is applied to the primary winding, the voltage at secondary winding will be:
- A) 2000V
 - B) 1000V
 - C) 500V
 - D) 100V
- Q19. *** A transformer with primary winding having 500 turns, secondary winding having 100 turns, if 200 A is flowing into the primary winding, the current out of the secondary winding will be:
- A) 40 A
 - B) 100 A
 - C) 500 A
 - D) 1000 A
- Q20. *** A transformer with primary winding having 100 turns and secondary winding having 500 turns. If 200 A is flowing into the primary winding, the current out of the secondary winding will be:
- A) 40 A
 - B) 100 A
 - C) 500 A
 - D) 1000 A

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- Q21.** * A single-phase transformer has 1000 turns on the secondary and 100 turns on the primary. If the transformer is connected to 110-volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load current is:
- A) 2.727 A
 - B) 27.27 A
 - C) 272.7 A
 - D) none of the above
- Q22.** * A single phase transformer has 110 turns on the secondary and 1000 turns on primary. If the transformer is connected to 110 volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load current is:
- A) 330 A
 - B) 247.9 A
 - C) 272.7 A
 - D) None of the previous
- Q23.** * A single-phase transformer is connected to 110 volt and supply 11 volt with 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load impedance as referred to the primary is:
- A) 0.0004 Ω
 - B) 4.0 Ω
 - C) 4044 Ω
 - D) 403 Ω
- Q24.** * A single phase transformer has 110 turns on the secondary and 1000 turns on primary. If the transformer is connected to 110 volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load impedance as referred to the primary is:
- A) 0.04 Ω
 - B) 4 Ω
 - C) 3.96 Ω
 - D) 403 Ω

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- Q25. * A single phase transformer with a ratio of 10:10, if the primary has a torque of 50, then the secondary will generate:
- A) 50 V
 - B) 100 V
 - C) 25 V
 - D) None of the above
- Q26. * A single phase transformer with a ratio of 10:20, if the primary has a torque of 50, then the secondary will generate:
- A) 50
 - B) 100
 - C) 75
 - D) None of the above
- Q27. * The power losses in Real Transformers are:
- A) 5%
 - B) 25%
 - C) 50%
 - D) Different from one transformer to other
 - E) None of the above
- Q28. * The friction losses in Real Transformers are:
- A) 5%
 - B) 25%
 - C) 50%
 - D) Different from one transformer to other
 - E) None of the above
- Q29. * A 10 kVA 480/120 V auto-transformer is to be used to supply power from a 600 V source to a 480 V load. Consider the transformer to be ideal with output power 50 kVA, the power rating is:
- A) 50
 - B) 1.25
 - C) 5
 - D) None of the above

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Q30. * Which parameter changes during transformation action in transformer?

- A) Current
- B) Voltage
- C) Frequency
- D) both A and B

Q31. * Which parameter does not change during transformation action in transformer?

- A) Current
- B) Voltage
- C) Frequency
- D) both A and B

Q32. * For time varying currents, the field or waves will be:

- A) Magneto static
- B) Electrical
- C) Electromagnetic
- D) Electrostatic

Q33. * If the supply frequency to the transformer is increased, the iron loss will

- A) Increase
- B) Not change
- C) Decrease
- D) Cannot be determined

Q34. * If the supply frequency of a transformer increases then output voltage of the transformer will:

- A) Decrease
- B) Increase proportionately
- C) Remain same
- D) Increase very rapidly

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Q35. * SC test gives:

- A) Parallel parameters of equivalent circuit
- B) Neither series nor parallel parameter of equivalent circuit
- C) Series parameters of equivalent circuit
- D) Both parameters of equivalent circuit

Q36. * Short circuit test on transformers is conducted to determine:

- A) Copper losses
- B) Core losses
- C) Eddy current losses
- D) Hysteresis losses

Q37. * Why OC test is performed on LV side?

- A) It will not give losses if conducted on HV side
- B) Less voltage is required and parameters can be transformed to HV side
- C) HV side does not have connections for voltage
- D) Simple construction

Q38. In no-load test we short the secondary terminals.

- A) True
- B) False

Q39. * The open circuit test and short circuit test are performed on a transformer to determine the:

- A) Equivalent circuit of transformer
- B) Voltage regulation of transformer
- C) Efficiency of transformer
- D) All the above

Q40. * On open circuit transformer test, voltage at rated frequency is applied to:

- A) HV side
- B) LV side
- C) LV and HV side
- D) LV or HV side

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- Q41. *** In OC test all the power supplied is utilized for:
- A) Cannot be determined
 - B) Core losses
 - C) Iron losses
 - D) Windage losses
- Q42.** Open circuit test on transformers is conducted so as to get:
- A) Hysteresis losses
 - B) Copper losses
 - C) Core losses
 - D) Eddy current losses
- Q43. *** In a given transformer for given applied voltage, which of the following losses remain constant irrespective of load changes?
- A) Copper losses
 - B) Cannot be determined
 - C) Friction and windage losses
 - D) Hysteresis and eddy current losses
- Q44. *** In no load test we keep secondary terminals:
- A) Shorted
 - B) Shorted via fixed resistor
 - C) Open
 - D) Shorted via variable resistors
- Q45. *** Transformer open circuited test, input power consists of core losses in transformer.
- A) False
 - B) True
- Q46.** The power factor of a transformer on no load is poor due to:
- A) Magnetizing reactance of the transformer
 - B) Open-circuited secondary
 - C) Low primary winding resistance
 - D) Low no-load current

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Q47. * Transformer core is designed to reduce:

- A) Eddy current loss
- B) Hysteresis loss
- C) Hysteresis loss and eddy current loss
- D) Cannot be determined

Q48. * Transformer uses laminated core in order to:

- A) Reduce eddy current loss
- B) Reduce eddy current and hysteresis loss
- C) Ensure good magnetic coupling
- D) All of these

Q49. * Which of the following losses does not vary with load in transformer?

- A) Copper loss
- B) Hysteresis loss
- C) None of the above
- D) All losses in transformer vary with load

Q50. * Which of the following losses varies with load in transformer?

- A) Copper loss
- B) Hysteresis loss
- C) Eddy current loss
- D) All losses in transformer vary with load

Q51. * Which of the following loss varies with the load:

- A) Copper loss
- B) Hysteresis loss
- C) Eddy current loss
- D) All losses in transformer vary with load

Q52. Iron loss in transformer can be determined by:

- A) Short circuit test
- B) Open circuit test
- C) Both short circuit and open circuit tests
- D) None of the above

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- Q53.** * Point out the wrong statement. Magnetic leakage is undesirable in electrical machine because it:
- A) Lower their power efficiency
 - B) Lead to their increased weight
 - C) Increase their cost of manufacture
 - D) Produce fringing
- Q54.** In the power systems, the auto transformer is preferred over the ordinary two winding transformer due to:
- A) Safety of operation
 - B) Availability of taps
 - C) Material saving
 - D) All the mentioned
- Q55.** * The magnetic flux through a coil having a single turn is varying according to $\phi = 25t^2 + 10t + 5$ Wb. The induced voltage between the coil terminals at $t = 1$ s is:
- A) 10 V
 - B) 25 V
 - C) 50 V
 - D) 60 V
- Q56.** The magnetic reluctance of a core of ferromagnetic material with relative of 4000, a mean path length of 100cm and the cross-sectional area of 150cm^2 is:
- A) 0.01667 A.turn/Wb
 - B) 2500 A.turn/Wb
 - C) 8750 A.turn/Wb
 - D) 13263 A.turn/Wb
- Q57.** A core of ferromagnetic material has a cross-sectional area of 100cm^2 , if the magnetic flux through this core is 0.01Wb, the magnetic flux density within this core is:
- A) 0.1T
 - B) 100T
 - C) 10T
 - D) 1.0T
- Q58.** If the magnetic flux through a 100-turn coil increases from 0 to 0.025 Wb, the induced voltage between the ends of the coil is:
- A) 25 V
 - B) 100 V
 - C) 34 V
 - D) 2.5 V

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Q59. * The property of material which oppose the creation of magnetic flux in it is:

- A) Reluctivity
- B) Permeance
- C) Reluctance
- D) Magnetomotive force

Q60. * The copper of transformer must have:

- A) Low reluctance
- B) High resistance
- C) High reluctance
- D) Low resistance

Q61. * The core of transformer must have:

- A) low reluctance
- B) high resistance
- C) high reluctance
- D) low resistance

Q62. * The full-load copper loss of a transformer is 4000W, at half load, the copper loss will be:

- A) 400W
- B) 2000W
- C) 1000W
- D) 500W

Q63. * The full-load copper loss of a transformer is 2000W, at half load, the copper loss will be:

- A) 400W
- B) 2000W
- C) 1000W
- D) 500W

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Q64. * The working principle of transformer depends upon:

- A) Ohm's law
- B) Lenz's law
- C) Fleming's left hand rule
- D) Faraday's law of electromagnetic induction

Q65. * The law that describes the generated voltage by a varying magnetic field is:

- A) Gauss's law
- B) Ampere's law
- C) Ohm's law
- D) None of the above

Q66. * Transformer Parameters can be determined through:

- A) Faradays law
- B) Open circuit test and short circuit test
- C) Y- Δ connection set-up
- D) All the above

Q67. * The law describing the electromotive force induced in a conductor placed in varying magnetic field is:

- A) Ohm's law
- B) Gauss's law
- C) Faraday's law
- D) Ampere's law

Q68. * Transformers windings are generally made of:

- A) Iron
- B) Steel
- C) Steel iron alloy
- D) Copper

Q69. Whenever a conductor is placed in a varying magnetic field, an electromotive force is induced by Lenz law:

- A) True
- B) False

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Q70. * An ideal transformer is one which has:

- A) No winding resistance
- B) No leakage reactance
- C) No losses
- D) All of the above

Q71. * For an ideal transformer:

- A) Input power equal to output power
- B) Input power equal to output power and copper losses
- C) Input power equal to output power and hysteresis losses
- D) Both B and C

Q72. * The unit of Magneto-motive force is Ampere-wb.

- A) True
- B) False

Q73. * The unit of the magnetic flux is:

- A) Henry
- B) Weber
- C) Tesla
- D) Ampere/meter

Q74. * Induced EMF in the primary of transformer is equal to terminal voltage applied at primary:

- A) True
- B) False

Q75. * The induced voltage will oppose the flux producing it.

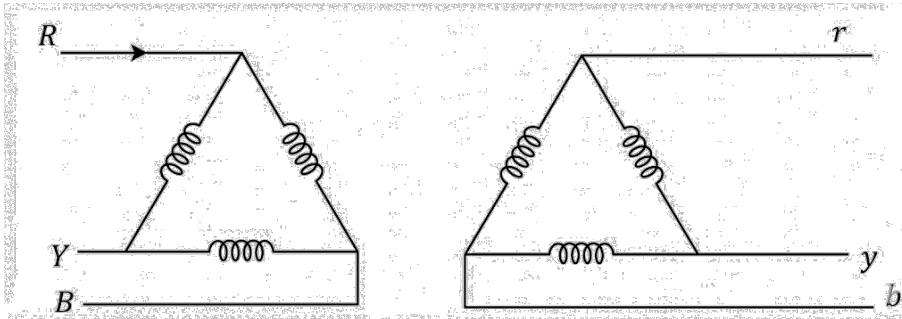
- A) True
- B) False

Q76. The induced voltage will not oppose the flux producing it.

- A) True
- B) False

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Q77. Below is the circuit diagram for delta-delta transformer have a ratio k . The current flowing in the windings of phase r-y in the below diagrams is:



- A) $k * i / 1.73$
- B) $i / k * 1.73$
- C) $1.37ki$
- D) $3ki$

Q78. If one of the transformers is removed from the bank of only delta-delta, then it will behave as power delivery transformer of:

- A) 58%
- B) 78%
- C) 45%
- D) 100%

Q79. * A 3-phase transformer has its primary connected in delta and secondary in star, the secondary to primary turns ratio per phase is 5, What would be the secondary voltage if the primary voltage is 400V:

- A) 80V
- B) 3364V
- C) 2000V
- D) 138V

Q80. * How 3-phase transformers are constructed:

- A) A bank of 3 single phase transformers
- B) A single 3-phase transformer with the primary and secondary of each phase wound on three legs of a common core
- C) Single 3-phase transformer or a bank of 3 single phase transformers
- D) By different method

Q81. A 3-phase transformer has a primary connected in Delta and secondary in Star. The line voltage ratio is:

- A) Equal to phase voltage ratio
- B) $\sqrt{3}$ times the phase voltage ratio
- C) $1/\sqrt{3}$ times the phase voltage ratio
- D) Independent of phase voltage ratio

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Q82. In three phase bank phases are:

- A) The phases are electrically not connected and magnetically independent
- B) The phases are electrically not connected and magnetically dependent
- C) The phases are electrically connected and magnetically independent
- D) The phases are electrically connected and magnetically dependent

Q83. The main reason for using a Δ connection on the primary side is:

- A) To reduce copper loss
- B) To eliminate third harmonic currents
- C) To increase voltage rating
- D) To increase current rating

Q84. * Three phase transformer compare to a bank of 3 single phase transformers is:

- A) Can't be determined
- B) Cheaper
- C) Costlier
- D) More in space

Q85. * The efficiencies of transformers compared to electric motors of the same power are:

- A) Much smaller
- B) About the same
- C) Much higher
- D) Can't comment

Q86. * The winding of the transformer that has less cross-sectional area of conductor is:

- A) Primary winding
- B) Secondary winding
- C) High voltage winding
- D) Low voltage winding

Q87. A transformer with large positive voltage regulation is an indication of:

- A) Leading load
- B) Resistive load
- C) lagging load
- D) could be any load

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Q88. * The relation between the direction of the force, acting on a current carrying conductor in a magnetic field, and the direction of magnetic field is:

- A) Unrelated
- B) Opposite direction
- C) The same direction
- D) Perpendicular

Q89. * The force exerted on a current-carrying wire placed in a magnetic field is maximum when the angle between the wire and the direction of magnetic field is:

- A) 45°
- B) 60°
- C) 90°
- D) 180°

Q90. * The basic function of power transformer is to transfer:

- A) DC to AC
- B) AC to DC
- C) AC to AC
- D) DC to DC

Q91. * The flux density can be controlled by:

- A) Current I only
- B) Material permeability only
- C) All the above
- D) None of the above

Q92. * As compared the transformer with other AC machines, it is more efficient because:

- A) It does not have mechanical losses
- B) It does not have copper losses
- C) Core losses are negligible
- D) None of the above

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Q93. Primary winding of a transformer:

- A) Is always a high voltage winding
- B) Is always a low voltage winding
- C) Could either be a low voltage or high voltage winding
- D) Cannot be determined

Q94. Secondary current of a step-down transformer is:

- A) Lower than primary current
- B) Higher than primary current
- C) Equal to primary current
- D) Double than primary current

Q95. * The primary and secondary windings of transformer are linked each other by:

- A) eddy current
- B) primary induction
- C) not linked at all
- D) None of the above

Q96. * The primary and secondary winding of transformer have linked each other by:

- A) Mutual conduction
- B) Mutual induction
- C) Not linked at all
- D) Self-induction

Q97. * A transformer transform:

- A) Current
- B) Voltage
- C) Power
- D) Frequency

Q98. Strength of the magnetic field is:

- A) Inversely proportional to the force applied
- B) Directly proportional to the force applied
- C) Directly proportional to the square of the force applied
- D) Inversely proportional to the square of the force applied

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Q99. * If a conductor is moving parallel to a uniform magnetic field, the voltage induced between the conductor terminals will:

- A) Be zero
- B) Depend of the intensity of the magnetic field
- C) Depend on the speed of the wire
- D) Be infinity

Q100. Wings present in an autotransformer are:

- A) 1
- B) 2
- C) 3
- D) 4

Q101. * A constant current flow in a horizontal wire in the plane of the paper from west to east. The direction of magnetic field at a point will be north to south:

- A) Directly above the wire
- B) Directly below the wire
- C) In the plane of the paper, in the north side of the wire
- D) In the plane of the paper, in the south side of the wire

Q102. * A constant current flows in a horizontal wire in the plane of the paper from east to west. The direction of magnetic field at a point will be north to south:

- A) Directly above the wire
- B) Directly below the wire
- C) In the plane of the paper, in the north side of the wire
- D) In the plane of the paper, in the south side of the wire

Q103. * 60 RPM is equal to ____ radian per second.

- A) 2π
- B) 60
- C) $2\pi / 60$
- D) $2\pi \times 60$

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- Q104.** * If the rated speed of a DC motor is 1200RPM and the rater torque is 40N.m, the rated power of this motor is:
- A) 30 kW
 - B) 5 kW
 - C) 3 kW
 - D) 0.5 kW
- Q105.** * The rated speed of a motor is 1800 RPM. If the rated torque of this motor is 140 N.m, the rated power of this motor is:
- A) 5 HP
 - B) 10 HP
 - C) 20 HP
 - D) 35 HP
- Q106.** * The rated power of a motor is 10 HP. If the rated torque of this motor is 40 N·m, the rated speed of this motor is:
- A) 1500 RPM
 - B) 2780 RPM
 - C) 800 RPM
 - D) none of the above
- Q107.** * The velocity of the wheel if the angular speed is 25 rad/s and radius is 10 m is:
- A) 250 m/s
 - B) 280 m/s
 - C) 270 m/s
 - D) 240 m/s
- Q108.** * A motor's shaft is spinning at a speed of 1800 r/min. What is the shaft speed in radians per second?
- A) 1800 rad/s
 - B) 188.5 rad/s
 - C) 11.41 rad/s
 - D) None of the above

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- Q109.** * A rotor of an electric machine starts rotation from rest, at constant acceleration of $9\pi \text{ rad/sec}^2$. The time required for the rotor to make one complete rotation is:
- A) $3/2 \text{ sec}$
 - B) $4/9 \text{ sec}$
 - C) $2/3 \text{ sec}$
 - D) $2/9 \text{ sec}$
- Q110.** * A rotor of an electric machine rotates at constant speed and makes 1500 revolutions per minute. The time required for one complete revolution in seconds is:
- A) 25
 - B) 1500
 - C) 2π
 - D) $1/25$
- Q111.** * A rotor of an electric machine rotates at constant speed and makes 2500 revolutions per minute required for one complete revolution in seconds is:
- A) 1500
 - B) $1/500$
 - C) 0.024
 - D) 0.04
- Q112.** * If the work done by a motor is 1000 J while its torque is 10 N.m, the number of complete revolutions done by the shaft of this motor is:
- A) 100π
 - B) 50π
 - C) $50/\pi$
 - D) $100/\pi$
- Q113.** * The work done by a motor shaft having a torque of 20 N.m when rotates one complete rotation is:
- A) $40/\pi \text{ J}$
 - B) $20\pi \text{ J}$
 - C) $40\pi \text{ J}$
 - D) $10/\pi \text{ J}$

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- Q114. *** A torque $\tau = 100 \text{ N}\cdot\text{m}$ is acting on a freely rotating shaft having a moment of inertia $J = 20 \text{ kg}\cdot\text{m}^2$. The angular acceleration of the shaft, caused by this torque is:
- A) 0.2 rad/sec^2
 - B) 50 rad/sec^2
 - C) 2000 rad/sec^2
 - D) 5 rad/sec^2
- Q115. *** A force $F = 10 \text{ N}$ is acting on a freely rotating shaft having a moment of inertia $J = 2 \text{ kg}\cdot\text{m}^2$, as shown in the figure. The angular acceleration of the shaft, caused by this force is:
- A) 0.25 rad/sec^2
 - B) $50\sqrt{3}/2 \text{ rad/sec}^2$
 - C) 0.5 rad/sec^2
 - D) $5\sqrt{3}/2 \text{ rad/sec}^2$
- Q116. *** The electromotive force induced in a conductor of length l moving at velocity v in a magnetic field having flux density B is:
- A) $e_{ind} = Blv$
 - B) $e_{ind} = B^2lv$
 - C) $e_{ind} = (B/l)v$
 - D) $e_{ind} = (l \times B)v$
- Q117. *** The electromotive force induced in a conductor of length l moving at velocity v in a magnetic field having a flux density B is:
- A) $e_{ind} = (l \times B) \times V$
 - B) $e_{ind} = (V \times B) \times l$
 - C) $e_{ind} = (l \times B) \cdot V$
 - D) $e_{ind} = (V \times B) \cdot l$
- Q118. *** The electromotive force induced in a conductor of length l moving at velocity v in a magnetic field having a flux density B is:
- A) $e_{ind} = (V \times B) \times l$
 - B) $e_{ind} = (V \cdot B) \cdot l$
 - C) $e_{ind} = (V \cdot B) \times l$
 - D) $e_{ind} = (V \times B) \cdot l$

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Q119. * The force F on a current carrying conductor of length l and current I placed perpendicular in a uniform magnetic field of flux density B is:

- A) $F = I/B$
- B) $F = B/I$
- C) $F = BIl$
- D) $F = IBl^2$

Q120. * The force F on a current carrying conductor of length l and current I , placed perpendicular in a uniform magnetic field of flux density B is:

- A) $F = l/BI$
- B) $F = B/II$
- C) $F = BII$
- D) $F = I/BI$

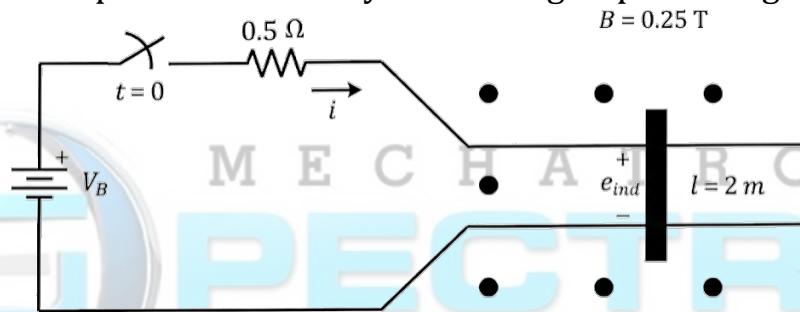
Q	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	A	B	B	A	C	D	A
Q	11	12	13	14	15	16	17	18	19	20
Answer	B	A	C	A	B	D	D	B	D	A
Q	21	22	23	24	25	26	27	28	29	30
Answer	A	B	B	B	D	D	D	E	C	D
Q	31	32	33	34	35	36	37	38	39	40
Answer	C	C	B	C	C	A	B	B	D	B
Q	41	42	43	44	45	46	47	48	49	50
Answer	B	C	D	C	B	A	C	B	B	A
Q	51	52	53	54	55	56	57	58	59	60
Answer	A	B	D	C	D	D	D	D	C	D
Q	61	62	63	64	65	66	67	68	69	70
Answer	A	C	D	D	D	B	C	D	B	D
Q	71	72	73	74	75	76	77	78	79	80
Answer	A	B	B	A	A	B	A	A	B	C
Q	81	82	83	84	85	86	87	88	89	90
Answer	B	C	B	B	C	C	C	D	C	C
Q	91	92	93	94	95	96	97	98	99	100
Answer	C	A	C	B	D	B	C	B	A	A
Q	101	102	103	104	105	106	107	108	109	110
Answer	A	B	A	B	D	D	A	B	C	D
Q	111	112	113	114	115	116	117	118	119	120
Answer	C	C	C	D	D	A	D	D	C	C

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Part. 2 (MID Problems):

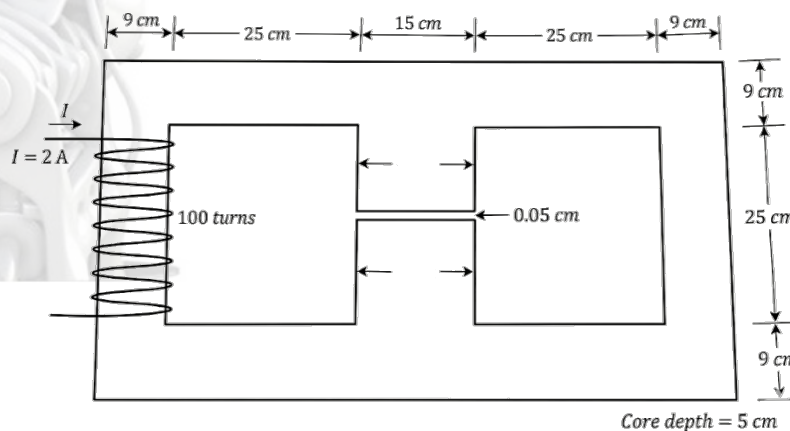
Q1. Consider the linear DC machine as shown, with $R = 0.5 \Omega$, $B = 0.25 \text{ T}$, and $L = 2 \text{ m}$, with no load steady state velocity of 300 m/s .

- A) Find e_{ind} and V_B .
- B) Direction of the bar movement at no load.
- C) I_{max} .
- D) Magnitude of external force required to charge battery with 5 volt .
- E) In (D), how much power the battery consuming or producing.



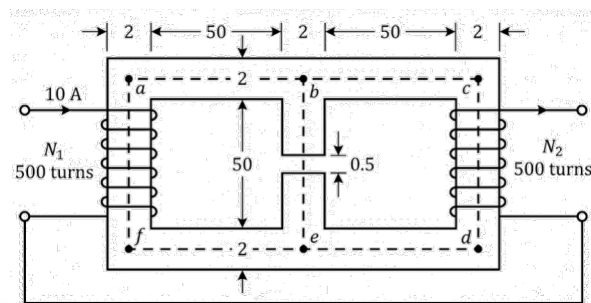
Q2. A core with three legs is shown. Its depth is 5 cm and there are 100 turns on the leftmost leg. The relative permeability of the core can be assumed to be 2000 and constant:

- A) What flux exist in each of the three legs of the core?
- B) What is the flux density in each of the legs?
- C) Assume a percent increase in the effective area of the air gap due to fringing effects.



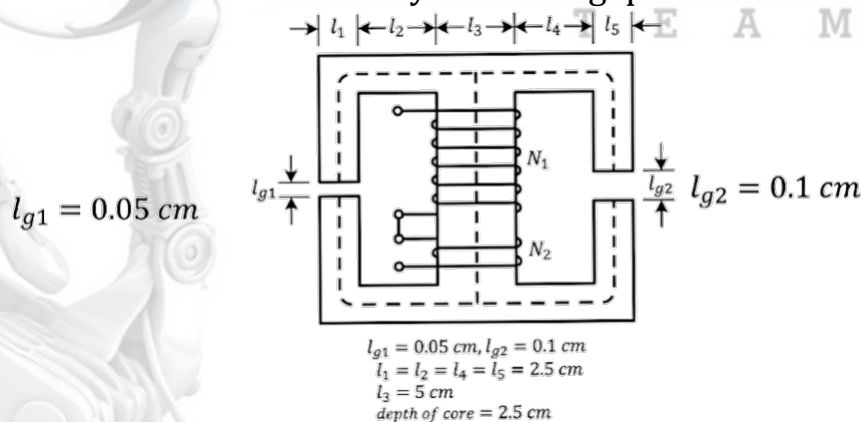
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Q3. A magnetic circuit as shown below with relative permeability of the ferromagnetic material is 1200. Neglect magnetic leakage and fringing. All dimensions are symmetric in centimeters and magnetic material has square cross section area. If the length for b-e only is 51.5 cm, find the airgap flux, airgap flux density, and magnetic field intensity in the air gap.



Q4. In the magnetic circuit shown below provides flux in the two air gaps. The coils $N_1 = 700$, $N_2 = 200$ are connected in series and carry current of 0.5 amperes. Neglect leakage flux, reluctance of the iron infinite permeability and fringing at the air gaps.

- Draw the equivalent circuit
- Determine the flux and flux density in the air gaps



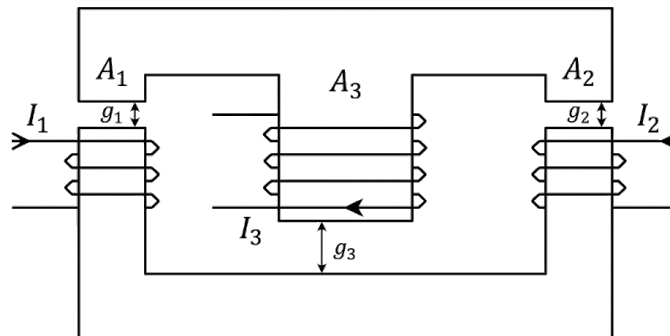
Q5. The magnetic circuit shown in Figure 2 has an infinitely permeable magnetic core. The parameters of the magnetic circuit are:

$$g_1 = 5 \text{ mm}, g_2 = 5 \text{ mm}, g_3 = 10 \text{ mm}$$

$$A_1 = 5 \text{ cm}^2, A_2 = 5 \text{ cm}^2, A_3 = 10 \text{ cm}^2$$

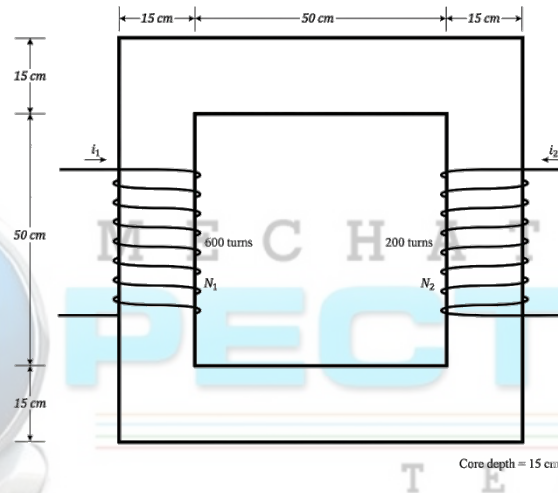
$$N_1 = 80 \text{ turns}, N_2 = 100 \text{ turns}, N_3 = 125 \text{ turns}.$$

A current of 12 A flows in the first coil, N_1 . The other coils are unexcited. Determine the flux and flux densities in each of the air gaps g_1 , g_2 and g_3 .



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- Q6. * A two-legged core is shown in the figure. The winding on the left leg of the core (N_1) has 600 turns, and the winding on the right (N_2) has 200 turns. The coils are wound in the directions shown in the figure. If the dimensions are as shown, then what flux would be produced by currents $i_1 = 5.0$ A and $i_2 = 1.0$ A? Assume $\mu_r = 1200$ and constant.

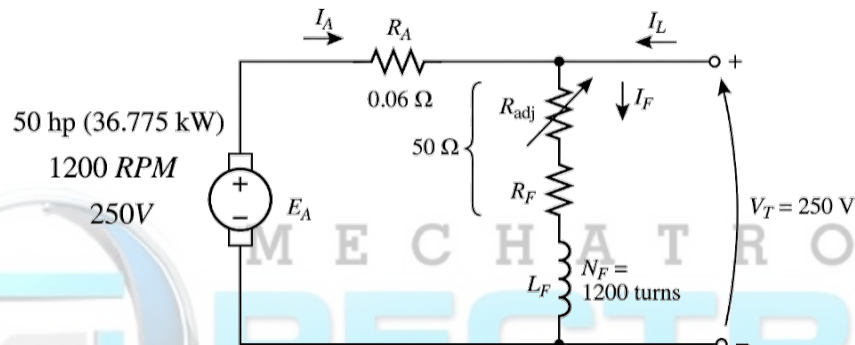


- Q7. A single phase transformer has 100 turns on the secondary and 1000 turns on primary. If transformer is connected to 110 volt supply while 3 KVA load connected to the secondary winding. Assuming ideal transformer, find:
- A) load impedance
 - B) load voltage
 - C) load impedance as referred to the primary
- Q8. For autotransformer:
- A) What are the uses of an autotransformer?
 - B) What is the difference between a power transformer and an autotransformer?
 - C) Draw the step-down autotransformer circuit.
 - D) Find the relationship between low current and high current.

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Part. 3 (FINAL MCQs):

For Q1. to Q5, The figure below shows the diagram of a 50 hp 36.775 kW, 250 V, 1200 RPM DC shunt motor with compensating windings and an armature resistance of $0.06\ \Omega$. The field circuit of the motor has a total resistance $R_{adj} + R_f = 50\ \Omega$ which produces a no load speed of 1200 RPM. There are 1200 turns per pole on the shunt field winding



Q1. The rated current of this motor is:

- A) 147.1 A
- B) 199.5 A
- C) 248.6 A
- D) 575.0 A

Q2. The rated torque of this motor is:

- A) 585.3 N · m
- B) 379.5 N · m
- C) 292.7 N · m
- D) 146.3 N · m

Q3. The speed of this motor when the line current is 100 A:

- A) 987RPM
- B) 1020RPM
- C) 1157RPM
- D) 1173RPM

Q4. The speed of this motor at rated current is:

- A) 987RPM
- B) 1020RPM
- C) 1157RPM
- D) 1173RPM

Q5. The speed regulation of this motor is:

- A) 2.78%
- B) 3.66%
- C) 5.67%
- D) 6.75%

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For Q6. to Q10, A shunt generator delivers 450 A at 230 V and the resistance of the shunt field and armature are $50\ \Omega$ and $0.03\ \Omega$ respectively.

Q6. The current through shunt field winding I_f is:

- A) 4.6 A
- B) 480 A
- C) 450 A
- D) None of the above

Q7. The load current is:

- A) 450 A
- B) 4.6 A
- C) 480 A
- D) None of the above

Q8. The armature current is:

- A) 454.6 A
- B) 480 A
- C) 450 A
- D) None of the above

Q9. The armature voltage drop is:

- A) 13.6 V
- B) 22700 V
- C) 0.138 V
- D) None of the above

Q10. The generated EMF is:

- A) 243.6 V
- B) 230.1 V
- C) 22930 V
- D) None of the above

M E C H A T R O N I C S
S P E C T R U M
T E A M

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Q11. * The torque in a DC shunt motor is proportional to:

- A) Armature current
- B) Field current only
- C) Square of armature current
- D) Armature resistance
- E) None of the above

Q12. * The armature torque of the DC shunt motor is proportional to:

- A) Field flux only
- B) Armature current only
- C) Field flux and armature current
- D) Field current

Q13. In a DC motor, the electromagnetic torque developed is:

- A) In opposite direction to both load torque and frictional torque
- B) In the same direction to the load torque but opposite direction to the frictional torque
- C) In the same direction as the load torque and frictional torque
- D) In the same direction as the frictional torque but in opposite direction to the load torque

Q14. * EMF and torque produced in a DC machine are proportional to ____ and ____ respectively

- A) Armature speed and armature EMF
- B) Armature EMF and armature speed
- C) Armature current and armature EMF
- D) Armature speed and armature current

Q15. * In DC motor, speed is

- A) Proportional to flux
- B) Proportional to square flux
- C) Inversely proportional to flux
- D) None of the above

Q16. In DC series machine ϕ is directly proportional to I_A not I_f .

- A) True
- B) False

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- Q17.** In DC machines the plane passing through the axis of the armature and through center of contact of the brushes is known as ____ plane.
- A) Middle
 - B) Central
 - C) Excitation
 - D) None of the above
- Q18.** The component used to produce a DC voltage output rather than AC voltage output is called:
- A) Slip ring
 - B) Brushes
 - C) Armature
 - D) Commutator
- Q19.** Slip ring is used to produce a DC voltage output rather than an AC voltage output.
- A) True
 - B) False
- Q20. *** If the field winding of a lightly loaded DC shunt motor is opened, the:
- A) Motor will stop
 - B) Motor speed will decrease slightly
 - C) Motor speed will increase slightly
 - D) Motor will pick up very high speed
- Q21. *** If the field winding of a DC shunt motor opens during operation, the motor will:
- A) Stop immediately
 - B) Run at dangerously high speed
 - C) Run at constant speed
 - D) Run at half speed
 - E) None of the above
- Q22. *** A DC shunt motor is running at light load, what happens if the field winding gets opened?
- A) Motor will stop
 - B) Motor will burn
 - C) Motor will make noise
 - D) Motor will pickup high speed

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- Q23.** If a DC shunt motor is working at full load and if shunt field circuit suddenly opens:
- A) Will make armature to take heavy current possibly burning it
 - B) Will result in excessive speed possibly destroying armature due to excessive centrifugal stresses
 - C) Nothing will happen to motor
 - D) Motor will come to stop
- Q24.** * If the load of the shunt DC motor is reduced, its speed will:
- A) Increase abruptly
 - B) Increase in proportion to speed reduction
 - C) Decrease in proportion to speed reduction
 - D) Remain almost constant
- Q25.** As the load is increased the speed of DC shunt motor will:
- A) Reduce slightly
 - B) Increase slightly
 - C) Increase proportionately
 - D) None of the above
- Q26.** The torque in a DC shunt motor is proportional to:
- A) Armature current
 - B) Field current only
 - C) Square of armature current
 - D) Armature resistance
 - E) None of the above
- Q27.** The armature torque of the DC shunt motor is proportional to:
- A) Field flux only
 - B) Armature current only
 - C) Field flux and armature current
 - D) Field current
- Q28.** In a DC shunt motor, speed is related to armature current as:
- A) Directly proportional to the armature current
 - B) Proportional to the square of the current
 - C) Independent of armature current
 - D) Inversely proportional to the armature current
- Q29.** In a DC shunt motor for zero armature current we get speed:
- A) Zero and maximum
 - B) Zero and minimum
 - C) Non-zero and maximum
 - D) Doesn't depend on armature current

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- Q30.** * In the separately excited DC motor, the relation between motor speed n and the field current I_f is:
- A) n is directly proportional to I_f
 - B) n is directly proportional to the square of I_f
 - C) n is inversely proportional to I_f
 - D) n is inversely proportional to the square of I_f
- Q31.** * In the DC series machine, the magnetic MOST Sroportional to the:
- A) Square of the armature current
 - B) Armature current
 - C) Square root of the armature current
 - D) Cube of the armature current
- Q32.** * The DC motor that is most suitable for producing high starting torque is:
- A) Separately excited
 - B) Shunt excited
 - C) Serially excited
 - D) Differentially compound
- Q33.** DC series motors are used:
- A) Where load is constant
 - B) Where load changes frequently
 - C) Where constant operating speed is needed
 - D) High starting speed
- Q34.** The speed-torque characteristic of a DC series motor is:
- A) Constant speed
 - B) Nearly constant torque
 - C) High starting torque, speed decreases with load
 - D) Linear between speed and torque
 - E) None of the above
- Q35.** For some percentage increase in the torque, which DC motor will have the least percentage increase of input current?
- A) Series motor
 - B) Shunt motor
 - C) Cumulative compound motor
 - D) Separately excited motor

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Q36. The material, typically used for brushes in DC machines is:

- A) Copper
- B) Carbon
- C) Silver
- D) Iron

Q37. In a compound DC motor, if the series field aids the shunt field, the motor is:

- A) Differential compound motor
- B) Cumulative compound motor
- C) Shunt motor
- D) None of the above

Q38. In shunt-excited DC motor, the relation between the speed and the armature current is:

- A) The speed is directly proportional to the armature current
- B) The speed is inversely proportional to the armature current
- C) The speed is inversely proportional to the square of the armature current
- D) The speed is directly proportional to the square of the armature current

Q39. Winding configuration for low voltage and high current is:

- A) Wave winding
- B) Lap winding
- C) Both (A) & (B)
- D) None of the above

Q40. The simple winding that has few parallel paths, regardless of the number of poles is called:

- A) Duplex winding
- B) Series winding
- C) Wave winding
- D) Lap winding

Q41. In a differential compound DC motor, in heavy loading condition:

- A) The torque increases rapidly
- B) The motor becomes unstable and may reverse
- C) The speed drops to zero
- D) The field current increases

Q42. Of the following, the type of the DC motor having the highest starting torque is:

- A) Shunt-excited
- B) Series motor
- C) Differentially compound excited
- D) None of the above

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Q43. * In the DC generator, the energy conversion is:

- A) From mechanical to electrical
- B) From electrical to mechanical
- C) From electrical to electrical
- D) From mechanical to mechanical

Q44. * In DC machine, mechanical losses are due to:

- A) Current
- B) Voltage
- C) None of the above
- D) All the above

Q45. If the field winding of a DC shunt generator is short-circuited, the output voltage will:

- A) Increase suddenly
- B) Remain the same
- C) Drop to zero
- D) Reverse in polarity

Q46. For self-excitation of a DC shunt generator, which condition is necessary?

- A) Field winding must have residual magnetism
- B) Field resistance $>$ critical resistance
- C) Load must be connected before starting
- D) External DC supply is mandatory

Q47. To reduce sparking, DC machines are provided with:

- A) Dummy coils
- B) Interpoles (commutating poles)
- C) Extra insulation between coils
- D) Series resistance in brushes
- E) Compensating winding
- F) None of the above

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- Q48. *** No-load speed of which of the following DC motor will be highest?
- A) Shunt motor
 - B) Series motor
 - C) None of the above
 - D) None of the above
- Q49. *** In order to reduce number of output voltage ripples in DC generator the following need to be implemented:
- A) Increase number of slip rings
 - B) Use shunt excitation
 - C) Use a pair of permanent magnetic pole
 - D) None of the above
- Q50. *** The speed of the DC shunt motor can be increased above its normal speed by:
- A) Increasing the field current
 - B) Decreasing the field current
 - C) Decreasing the terminal voltage
 - D) Increasing the armature resistance
- Q51.** The component used to produce DC voltage output rather than AC voltage output is:
- A) Slip ring
 - B) Brushes
 - C) Armature
 - D) Commutator
- Q52. *** What is armature reaction in a DC machine?
- A) The resistance offered by the armature winding
 - B) The heat generated in the armature
 - C) The effect of armature current on the main magnetic field
 - D) The reaction force experienced by the armature
- Q53. *** Why is armature reaction undesirable in DC machines?
- A) It causes distortion of the magnetic field and affects commutation
 - B) It improves the efficiency of the machine
 - C) It increases the size of the machine
 - D) It reduces the cost of the machine

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- Q54. *** What is the effect of armature reaction on the output voltage of a DC generator?
- A) It increases the output voltage
 - B) It has no effect on the output voltage
 - C) It decreases the output voltage
 - D) It stabilizes the output voltage
- Q55. *** What is the primary effect of armature reaction in a DC generator?
- A) It increases the speed of the generator
 - B) It increases the terminal voltage
 - C) It reduces the efficiency of the machine
 - D) It distorts the main magnetic field and shifts the neutral plane
- Q56. *** What happens to the neutral plane in a DC machine due to armature reaction?
- A) It shifts opposite to the direction of rotation in a motor
 - B) It shifts along the direction of rotation in a generator
 - C) Both (B) and (C)
 - D) It remains stationary
- Q57. *** How does armature reaction affect the performance of a DC motor?
- A) It decreases the speed of the motor
 - B) It causes excessive heating in the armature
 - C) It reduces torque and efficiency
 - D) It improves commutation
- Q58. *** What is the purpose of interpoles in a DC machine?
- A) To generate the main magnetic field
 - B) To reduce sparking at the brushes caused by armature reaction
 - C) To regulate the armature current
 - D) To increase the speed of the machine
- Q59. *** Which of the following is a characteristic of progressive wave winding?
- A) It is mainly used for low-speed machines
 - B) It requires more commutator segments than retrogressive winding
 - C) It increases the number of parallel paths
 - D) Each coil advances in the same direction

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- Q60. *** The main advantage of wave winding over lap winding is:
- A) Fewer parallel paths, leading to higher voltage
 - B) Higher efficiency
 - C) Simpler design
 - D) Lower copper losses
- Q61. *** What happens if the polarity of the supply voltage to a DC motor is reversed?
- A) The motor stops rotating
 - B) The speed of the motor increases
 - C) The motor rotates in the opposite direction
 - D) The motor burns out
- Q62. *** The commutator pitch for the simplex retrogressive winding is:
- A) -2
 - B) -1
 - C) $+1$
 - D) $+2$
- Q63. *** Commutator Pitch for a duplex progressive winding is:
- A) 1
 - B) -2
 - C) +1
 - D) +2
- Q64.** The simplex winding that has two parallel current paths regardless of number of pole called:
- A) Duplex winding
 - B) Series winding
 - C) Wave winding
 - D) Lap winding
- Q65.** Compensating winding will provide incomplete neutralization:
- A) Under pole region
 - B) In interpole region
 - C) Everywhere
 - D) Complete neutralization

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Q66. Slip ring is used to produce a DC voltage output rather than an AC voltage output.

- A) True
- B) False

Q67. The horse power obtained from the motor shaft is called:

- A) Armature reaction
- B) Magnetomotive force
- C) Useful torque
- D) None of the above

Q68. * Which of the following statement is true:

- A) The armature is the rotor
- B) The armature is the stator
- C) Can be both
- D) Is the primary side

Q69. * Displacement is a _____ quantity.

- A) Scalar
- B) Vector
- C) Scalar and Vector
- D) Tensor

Q70. * Which of the following are effects of armature reaction?

- A) Increase in iron losses
- B) Commutation problems
- C) Additional commutation sparking
- D) All the above

Q71. In machines the plane passing through the axis of the armature and through center of contact of the brushes is known as:

- A) MNA
- B) Neutral axis
- C) GNA
- D) None of the above

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- Q72.** For a DC machine, if internal generated voltage E_A of 167 V at a speed 1200 r/min, then the speed for induced voltage 240 V would be:
- A) 1195 r/min
 - B) 1835 r/min
 - C) 1725 r/min
 - D) None of the above
- Q73. *** If a DC motor has a full-load speed of 750 RPM & SR of 15%, the no-load speed is:
- A) 862.5 RPM
 - B) 800.5 RPM
 - C) 750 RPM
 - D) None of the above
- Q74. *** A DC shunt generator delivers 20 A at 220 V. Armature resistance is $0.4\ \Omega$ and shunt field resistance is $110\ \Omega$. The generated EMF is:
- A) 224 V
 - B) 228 V
 - C) 230 V
 - D) 235 V
- Q75.** A DC shunt motor has a flux per pole of 0.03 Wb and armature current of 20 A. If the torque constant is unity, the developed torque is:
- A) 0.6 N·m
 - B) 1.2 N·m
 - C) 2.4 N·m
 - D) 6.0 N·m
- Q76.** A DC motor develops 10 kW at 1000 RPM. The developed torque is approximately:
- A) 47.7 N·m
 - B) 95.5 N·m
 - C) 143 N·m
 - D) 191 N·m
- Q77.** A DC generator supplies 50 A at 240 V. If total losses are 2 kW, the efficiency is:
- A) 80%
 - B) 85%
 - C) 88%
 - D) 92%

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Q78. * What is the E generated in a 4 pole DC machine having flux per pole equal to 0.1 wb rotating at 1500 RPM? Number of coil sides 100 and K' is 127:

- A) 19 kV
- B) 1.9 kV
- C) 190 V
- D) 19 V

Q79. * A 4-pole DC generator has a wave-wound armature with 720 conductors. If the flux per pole is 0.025 Wb and speed is 600 RPM, the generated EMF is:

- A) 150 V
- B) 180 V
- C) 216 V
- D) 240 V

Q80. For an 8-pole, 25 kW, 120 V DC generator has a duplex lap-wound armature which has 64 coils with 10 turns per coil. Its rated speed is 3600 r/min, number of current paths is:

- A) 16
- B) 8
- C) 24
- D) None of the above

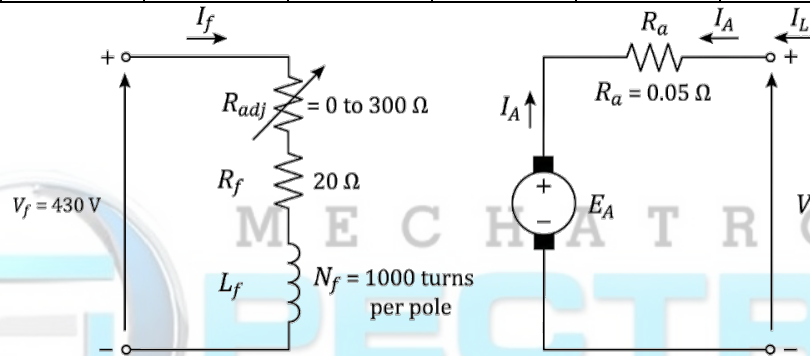
Q	1	2	3	4	5	6	7	8	9	10
Answer	A	C	D	C	B	A	A	A	A	A
Q	11	12	13	14	15	16	17	18	19	20
Answer	A	C	A	D	C	A	A	D	B	D
Q	21	22	23	24	25	26	27	28	29	30
Answer	D	D	B	D	A	A	C	C	C	C
Q	31	32	33	34	35	36	37	38	39	40
Answer	A	C	D	C	A	B	B	B	B	C
Q	41	42	43	44	45	46	47	48	49	50
Answer	B	B	A	C	C	A	B	B	D	B
Q	51	52	53	44	45	46	47	48	49	60
Answer	D	C	A	C	D	C	C	B	D	A
Q	61	62	63	64	65	66	67	68	69	70
Answer	C	B	D	C	B	B	C	A	B	D
Q	71	72	73	74	75	76	77	78	79	80
Answer	A	C	A	B	A	B	B	A	B	A

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Part. 4 (FINAL Problems):

Q1. * A separately excited DC motor is rated at 172 kW, 430 V, 400 A, and 1800 r/min. and its magnetization curve is shown below. When $R_{adj} = 63\Omega$.

I_f	0.0	1.0	2.0	3.0	4.0	4.75	5.0	5.2	6.0
V_T	20	120	220	320	390	420	430	435	460

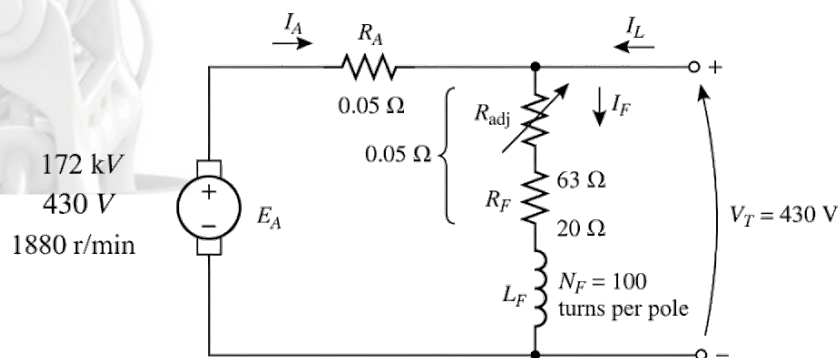


Find:

- A) I_f
- B) E_A
- C) τ

Q2. * A shunt DC motor is rated at 172 kW, 430 V, 400 A, and 1800 r/min. and its magnetization curve is shown below. When $R_{adj} = 63\Omega$.

I_f	0.0	1.0	2.0	3.0	4.0	4.75	5.0	5.2	6.0
E_A	0	100	200	300	370	400	410	415	440



Find:

- A) n_0
- B) n at full-load
- C) SR

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Q3. DC motor delivers 15 kW at 1000 RPM. Total losses are 2.5kW. Find:

- A) Input power
- B) Efficiency
- C) Developed torque

Q4. DC generator gives 250 V at 1000 RPM. If speed is increased by 20% and flux is reduced by 10%, find the new generated EMF.

Q5. * A 30 kW output power machine with full speed efficiency of 92%. If the machine is powered by 240 V DC source and full speed of 30 m/s, find the effective length of the “bar” as linear machine.

Q6. * A 500 V shunt DC motor operates at 1100 RPM and delivers an output power of 36.3 kW. The motor has an efficiency of 90%. The field current is 1.8 A. and the armature resistance is 0.24Ω . Find:

- A) Full load current
- B) Full load torque

Q7. Solve the following problem for simplex progressive Lap-winding:

A six pole DC generator has a flux per pole of 30 mWb. The armature has 536 conductors connected as a lap. The DC generator runs at 1050 rpm, and delivers a rated armature current of 225 A to a load connected to its terminals. The field and armature windings are connected in series $R_f = R_a = 0.05 \Omega$. Combined friction, windage, and core losses are 2 kW. Calculate:

- A) Generated voltage E_g
- B) The power developed by armature P_{dev}
- C) The developed torque τ_{dev}
- D) Terminal voltage V_T
- E) Efficiency